Variable Dual Time Scale Caching Update and Resource Allocation Scheme for UAV-Assisted Edge Networks

Taijun Peng, Chao Yang, Qiutian Xu, Xin Chen, and Yi Liu

***Abstract*— Leveraging wireless caching in uncrewed aerial vehicles (UAVs)-assisted edge networks, UAVs can precache part of popular contents and fly to the users directly, the repeated backhaul transmission delays are reduced. However, the limited caching space, complex network environments and varying user demands make a challenge for designing an effective caching content selection and update scheme. The caching update of UAVs introduces an additional cost, and UAVs can move to the suitable locations in real time to improve the air-to-ground transmission rates. In this article, we propose a variable dual time scale caching update and resource allocation (VD-CURA) scheme for UAV-assisted edge networks, to minimize the average system task processing latency per unit content size. A joint optimization problem is proposed that the caching update for UAVs, including the caching content selection and update interval are optimized at the large time scale, and the UAV trajectory optimization and bandwidth allocation are performed at each small scale. For the varying networks, a hierarchical reinforcement learning (HRL)-based VD-CURA framework with two layers is proposed, with combined proximal policy optimization (PPO) in the upper layer and deep deterministic policy gradient (DDPG) in the lower layer. Notably, we introduce a new caching content selection metric Experimental results demonstrate the robust convergence and adaptability of the proposed VD-CURA solution algorithm, compared with the baseline schemes, the average latency per unit of content is reduced significantly.**

***Index Terms*— Caching update, dual time scale, hierarchical reinforcement learning (RL), resource allocation, uncrewed aerial vehicles (UAVs).**

Received 4 January 2026; revised 25 April 2026; accepted 22 May 2026. Date of publication 29 May 2026; date of current version 26 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62320106008 and Grant 62003094 and in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515012745. *(Corresponding author: Chao Yang.)*

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Digital Object Identifier 10.1109/JMASS.2026.3698250

I. Introduction

WITH the rapid development of the low-altitude economy, uncrewed aerial vehicles (UAVs) can provide ubiquitous coverage and relay communication services for data dissemination and collection in various fields including intelligent transportation and emergency communications [1], [2]. Due to their flexible deployment, high maneuverability, and efficient communication capabilities, UAVs are employed dynamically to ensure the stable communication links, or perform as edge computing (EC) node to meet spatio-temporal variable users' demands [3], [4]. Wireless edge caching has shown great potential for avoiding duplicate data transmissions and reducing network cloud to end backhaul transmission latency. It can select the task processing location optimally, and the distributions of both the computation and communication resources are also changed. Integrating the edge caching techniques to UAV networks can alleviate congestion in the backhaul links. The limited energy and storage capacities of UAVs restrict the upper limits of the caching strategy performance [5], [6], [7], [8].

For the caching scheme in UAV-assisted edge networks, the ideal situation is that the UAV meets the user requirements immediately, which means that UAVs had cached the corresponding data and be closer to the users. The caching update, communication resource allocation and UAV flying trajectories should be considered. Given that the caching update introduces additional costs, such as the transmission cost and time consumption, the caching update interval in a larger time scale can enhance the content utilization and decreases the total update latency. However, when the update interval is too long, cached content becomes outdated and user demands are not met in a timely manner. The cache hit ratio declines ultimately. Meanwhile, the UAV moving and communication bandwidth allocation can be optimized in a shorter time scale, to improve the link communication quality, and the task processing latency is reduced [9], [10]. Currently, some existing studies assume that UAVs refresh cached content in a fixed manner, that the cached contents always be updated at the beginning of every fixed time

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frame [11], the performance also be poor under the various user demands or network environments.

Expect the caching update interval and frequency, the caching content selected also be the important factor. Some existing works employ simple statistics from past caching results to rank contents, such as: the system average latency or request counts [7], [12]. However, in the context of multiple consecutive time duration for complex tasks, a potential discrepancy arises. Some tasks may receive many user requests but have short content usage time durations, or entail repeated requests for other parts of the same content type. Although such contents experience frequent requests, its overall utilization remains low. When the caching decisions are made purely based on the number of requests, the cache overhead may also increase.

In this article, we propose a novel variable dual time scale caching update and content selecting scheme for an UAV-assisted edge network, the flexible deployment of UAVs and the cost of caching content update are jointly considered. In detail, the caching update interval, the caching content selection, the corresponding UAV flying trajectory and communication resource allocation are mainly optimized. The system operates in a time frame-by-frame fashion, and each frame is divided into a set of time slots. The cached content selection and update are optimized at the beginning of each frame, the UAV trajectory and bandwidth allocation is optimize at each slot, to meet the dynamic user requirements and the network transmission environments. The time frame duration is determined adaptively according to the content size and usable duration, we formulate a novel objective that minimizes the average delay per unit content size, enabling fair and normalized performance evaluation. Moreover, we introduce a novel cache priority metric that accounts for both content usage duration and user request frequency for other parts of the same content type. In each slot, UAVs can fly to users which had matched the cached contents, then the bandwidth is saved for the user to UAV links. The main contributions are summarized as follows.

1) We construct a variable dual time scale caching scheme for UAV-assisted edge networks. The caching content update and the movement of UAVs are optimized in different time scales to meet the dynamic network environment and user requests.
2) We design an optimal caching update interval based on content size and usable duration, and a novel objective is formulated to minimize the average delay per unit content size, enabling fair and normalized performance evaluation At each time frame, a new caching content selection metric is proposed.
3) We propose a hierarchical reinforcement learning (HRL)-based variable dual time scale caching update and resource allocation (VD-CURA) algorithm to solve the formulated joint optimization problem, the caching update interval, caching deployment, UAV trajectories and bandwidth allocation are optimized orderly.

The rest of this article is organized as follows. Section II reviews related work. Section III presents the system model. Section IV analyzes the formulated optimization problem, and Section V introduces the VD-CURA algorithm. Section VI provides numerical examples and performance evaluations. This article is concluded in Section VII.

II. Related Work

Wireless caching is an crucial topic in the UAV-assisted edge networks. The main research issues focus on cache content selection, update strategies and joint optimization of caching with other technologies.

Edge caching enhances UAV network performance by reducing latency and load [13], [14], [15]. Numerous studies have explored the integration of UAV networks with edge caching [16], [17], [18]. Zheng et al. [19] investigated computation service caching in multiple UAV-enabled EC systems by formulating a joint optimization problem for service caching, task offloading, resource allocation, and UAV placement. Muslih et al. [20] examined dynamic UAV caching networks, UAVs serves as aerial BSs serving ground users. Through shared caching and location adjustments, a DDPG approach optimizes UAV mobility, user association, and caching is proposed under varying cache capacities and user densities. In [21], an adaptive video caching scheme was proposed, via jointly optimizing UAV/BS caching decisions and UAV flight paths. A combined proximal policy optimization (PPO) and long short term memory (LSTM) approach were designed, to demonstrate the substantial performance improvements. In [22], an integrated solution merges edge caching and UAV communication in dense cellular networks was proposed, by jointly optimizing user association, cache placement, UAV trajectories and transmit powers.

Although these efforts combine UAVs and EC to improve system performance, the caching update incurs extra delay and overhead. In [23], a two time scale joint optimization method addresses task offloading, resource allocation, and UAV trajectory control was proposed, enhancing the overall system caching utility. Similarly, Lin et al. [24] proposed a double-timescale approach with an enhanced service caching policy and TD3, significantly reducing latency and satisfying energy limits. Chen et al. [25] proposed a two-timescale scheme with a hierarchical multiagent reinforcement learning (RL) framework to optimize UAV service caching, flight paths, and mobile user's offloading, reducing overall energy consumption. Similarly, Zheng et al. [19] integrated multi-UAV EC networks with edge caching, jointly optimizing caching, offloading, resource allocation and UAV deployments to minimize the maximum task delay under energy budgets. To address difficulties in video caching, Zhang et al. [15] proposed a DRL based algorithm to optimize caching and user association. In [26], the system throughput was improved by processing and caching video at BSs. To improve the cache hit ratios and reduce content access latency in vehicular networks, Wu et al. [27] integrated mobility prediction, consistent hashing, and federated learning into a novel cooperative caching scheme for vehicle EC.

In summary, the existing dual time scale approaches generally update cached content uniformly, failing to adjust intervals

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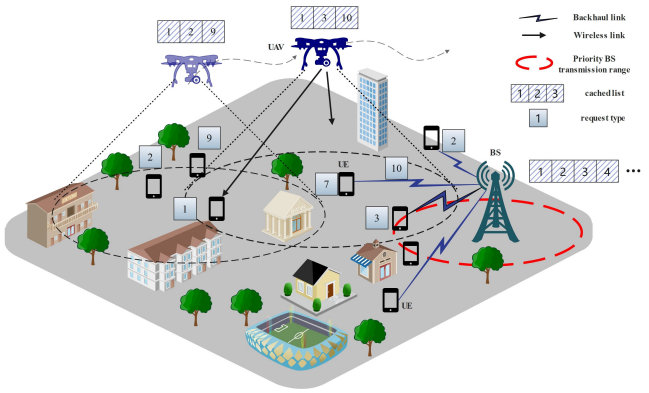


Fig. 1. System model.

based on specific content characteristics dynamically. This uniformity limits optimization gains in complex and real-time UAV-assisted network scenarios. To address this limitation, we propose an innovative variable dual time scale caching scheme enhanced by HRL, which determines the caching intervals based on content size and usage duration first, and the UAV trajectories and bandwidth allocation are jointly optimized to enhance the content delivery efficiency.

III. System Model

The system model of UAV-assisted edge networks is shown in Fig. 1, UAVs with cached contents assists BS to provide computing, communication resources for the ground densely users. Set that the BS can response all of the user requests, with the cloud computing center. The system operates in a time frame-by-frame fashion, and each frame includes a set of time slots. At time slot t, the 3-D coordinates of UAV u are denoted as $Q_u^t = (x_u^t, y_u^t, H)$, while the BS remains fixed at $Q_0 = (x_0, y_0, 0)$, $u \in \mathcal{U} = \{1, 2, \ldots, U\}$. UAVs are assumed to fly at a fixed altitude H [28], [29], and the trajectory optimization is performed in the horizontal 2-D plane, the moving trajectory is represented as $\mathbf{Q}_u = \{Q_u^1, \ldots, Q_u^k\}$, where k is the index of time slots in a frame. To ensure efficient and safe UAV operations, UAV trajectory is constrained to a feasible region $[Q_{Low}, Q_{Up}]$, and

$$\begin{cases} v_{\min}\Delta t \le \left\| Q_u^{t+1} - Q_u^t \right\| \le v_{\max}\Delta t \\ Q_{\text{Low}} \le Q_u \le Q_{Up} \end{cases} \tag{1}$$

where Δt is a given time slot interval, $v_{\max}$ and $v_{\min}$ are the thresholds of flight speeds. At slot t, user m generates a task f, characterized by the tuple $\langle l_{m,f}^t, D_{m,f}^t \rangle$. $m \in \mathcal{M} = \{1, 2, \ldots, M\}$. Here, $l_{m,f}^t$ denotes the data size of the requested content and $D_{m,f}^t$ is the transmission delay. Denote $\mathcal{F}$ as the set of tasks requested by user m, and each task is treated as an independent request. Since the tasks are content-oriented, individual tasks typically do not require significant computational resources [30]. In addition, the contents are assumed to be changed dynamically in each slot, but the total number of available content items is limited [31], [32].

The caching content deployment affects the task processing selection. When UAV possesses a request and the user is within its coverage area, the cached contents are transmitted to the user directly. Conversely, if the user is located outside the coverage zone, the request is forwarded to BS. For situations where the user is both near the BS and within the UAV's coverage area, the system prioritizes retrieving content from the BS. To enhance the efficiency of task execution, UAVs are capable of processing multiple user requests simultaneously. We set that the UAV has sufficient onboard power to complete the assigned tasks, it is consistent with the existing UAV-assisted network works that focus on communication and caching optimization without explicitly modeling propulsion energy [33], [34], the conclusion of this article can be extended to the case incorporating onboard energy constraints into the joint optimization framework directly. Define a binary decision variable $\rho_{m,f}^t \in \{0, 1\}$, where $\rho_{m,f}^t = 1$ signifies that the task f for user m at slot t is processed by UAV, and $\rho_{m,f}^t = 0$, the task is processed by BS. In this article, we focus on the caching deployment, when the UAV had cached the contents, the corresponding task can be processed at the UAV when the users are within the efficient communication range.

A. User Mobility Model

The high-order Gauss-Markov stochastic model (GMSM) is utilized to characterize user mobility [35], it is recognized with comparable speeds and directions, the velocity and direction of user m at time slot t are defined as

$$v_m^t = \kappa_1 v_m^{t-1} + (1 - \kappa_1)\,\bar{v} + \sqrt{1 - \kappa_1^2}\Phi_m \tag{2}$$

$$\theta_m^t = \kappa_2 \theta_m^{t-1} + (1 - \kappa_2)\,\bar{\theta} + \sqrt{1 - \kappa_2^2}\Psi_m \tag{3}$$

where κ_1 and κ_2 quantify the impact of the previous state on the current state, with both parameters confined to the range $(0, 1)$. $\bar{v}$ and $\bar{\theta}$ represent the user's mean speed and direction, respectively. The stochastic elements of the user's mobility are modeled by Φ_m and Ψ_m, following Gaussian distributions characterized by means $(\bar{\zeta}_v, \bar{\zeta}_\theta)$ and variances $(\sigma_v^2, \sigma_\theta^2)$. Then, the horizontal coordinates of user m are expressed as

$$x_m^t = x_m^{t-1} + v_m^t \tau \cos\left(\theta_m^{t-1}\right) \tag{4}$$

$$y_m^t = y_m^{t-1} + v_m^t \tau \sin\left(\theta_m^{t-1}\right). \tag{5}$$

B. Communication Model

Due to the utilization of orthogonal frequency-division multiple access (OFDMA) [36], [37], each user is allocated orthogonal resource blocks, and the interuser interference is disregarded effectively. At time slot t, the horizontal distance separating UAV u and user m is defined as

$$d_{m,u}^t = \sqrt{\left(x_m^t - x_u^t\right)^2 + \left(y_m^t - y_u^t\right)^2}. \tag{6}$$

Similarly, the distances between UAV u and ground BS s, user m and BS, are defined as

$$d_{u,s}^t = \sqrt{\left(x_u^t - x_0^t\right)^2 + \left(y_u^t - y_0^t\right)^2} \tag{7}$$

$$d_{m,s}^t = \sqrt{\left(x_m^t - x_0\right)^2 + \left(y_m^t - y_0\right)^2}. \tag{8}$$

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Either line-of-sight (LoS) and non-LoS (NLoS) are considered for the UAV communications. According to [38], the LoS probability between user m and UAV u is

$$P_{\mathrm{LoS}}\left(d_{m,u}^{t}\right)=\frac{1}{1+\beta_0\exp\left(-\beta_1\left(\frac{180}{\pi}\arcsin\left(\frac{h}{d_{m,u}^{t}}\right)-\beta_0\right)\right)} \tag{9}$$

where β_0 and β_1 are environment-dependent constants. Consequently, the NLoS probability is given as $P_{\mathrm{NLoS}}(d_{m,u}^{t}) = 1 - P_{\mathrm{LoS}}(d_{m,u}^{t})$. At time t, the path loss between user m and UAV u is calculated as

$$PL\left(d_{m,u}^{t}\right)=\begin{cases}\left(\frac{4\pi f_c d_{m,u}^{t}}{c}\right)^2\eta_{\mathrm{LoS}}, & \text{with } P_{\mathrm{LoS}}\left(d_{m,u}^{t}\right)\\ \left(\frac{4\pi f_c d_{m,u}^{t}}{c}\right)^2\eta_{\mathrm{NLoS}}, & \text{otherwise}\end{cases} \tag{10}$$

where f_c is the carrier frequency, and c is the speed of light. η_{LoS} and η_{NLoS} correspond to the additional path loss factors for LoS and NLoS links, respectively, following the widely adopted air-to-ground user (A2G) channel model in [39] and [40], and $\eta_{\mathrm{NLoS}} > \eta_{\mathrm{LoS}} > 1$.

Given that the user request is merely a content index signal with negligible data size [30], [41], the uplink transmission delay is ignored, and only the downlink content delivery delay is considered. Consequently, the downlink data transmission rate from UAV u to user m at slot t is

$$R_{m,u}^{t}=b_{m,u}^{t}W\log_2\left(1+\frac{\alpha\cdot p_u}{PL\left(d_{m,u}^{t}\right)\cdot\sigma^2}\right) \tag{11}$$

where α is the reference unit signal gain, W is the total system bandwidth, and p_u is the transmission power of UAV u, σ^2 denote the noise power. As the UAV–user and BS–user transmission links share the spectrum resources, the remaining spectrum can be allocated to the BS–user link in proportion to the UAV–user link's bandwidth sharing. Let $b_{m,u}^{t}, b_{m,s}^{t} \in (0, 1)$ represent the bandwidth resources allocated to user m at time t by UAV u or BS s, $\sum_{m=1}^{M}\sum_{u=1}^{U} b_{m,u}^{t} + b_{m,s}^{t} = 1$.

Considering the rain attenuation [42], the channel gain between the ground BS and user m at time t is

$$G_m^{t}=\frac{|g_m^{t}|^2}{\left(d_{m,s}^{t}\right)^{-a_3}} \tag{12}$$

where g_m^{t} is an independent, complex Gaussian random variable with zero mean and unit variance, and a_3 is the path loss exponent. For the downlink between BS and user m at time t [43], [44], the SNR is

$$\gamma_{m,s}^{t}=\frac{p_{0,m}^{t}|G_m^{t}|^2}{\sigma^2} \tag{13}$$

where $p_{0,m}^{t}$ is the transmission power, $b_{m,s}^{t}$ denotes the bandwidth resources allocated to user m, and N_0 is the noise power spectral density. Then, the data transmission rate between BS and user m is expressed as

$$R_{m,s}^{t}=b_{m,s}^{t}W\log_2\left(1+\gamma_{m,s}^{t}\right). \tag{14}$$

In addition, the transmission rate from BS to UAV u is

$$R_{u,s}^{t}=b_{u,s}^{t}W\log_2\left(1+\frac{\alpha\cdot p_s}{PL\left(d_{s,u}^{t}\right)\cdot\sigma^2}\right) \tag{15}$$

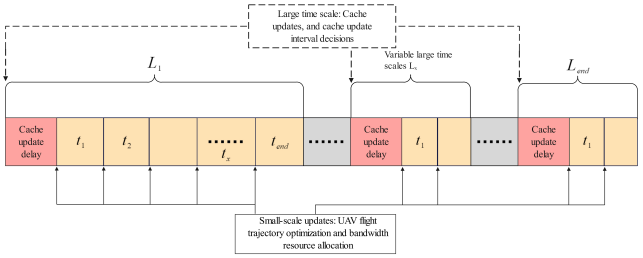


Fig. 2. Dual time scale caching update scheme.

where p_s is the transmit power of BS. It is worth noting that we assume the UAVs remain stationary during cache updates, with the entire available bandwidth is allocated to the UAVs to BS links, $\sum_{u=1}^{U} b_{u,s}^{t} = 1$.

C. Dual Time Scale Caching Update

Content caching updates incur a certain delay. In the system model, UAVs receive the contents from BS, and a delay is introduced during the update process. Therefore, we design a dual time scale caching update and UAV trajectory optimization scheme in UAV-assisted EC networks, as shown in Fig. 2. Caching update is performs at the beginning of each frame, and the UAV moving and resource allocation are optimized in each slot. During the time frame, cached contents remain unchanged until the next update interval is reached. Specially, the lengths of time frames $\{L_1, \ldots, L_n, \ldots, L_N\}$ are optimized to satisfy the dynamic requests of users.

For caching update model, we design a novel content caching priority metric, which consists of three parts.

1) The number $N_{m,f}^{d}$ of different users requests for content f. A larger value of $N_{m,f}^{d}$ indicates that the content is more popular.
2) The total duration T_f^{L} that multiple users continuously use content f, it includes cases where the content is used simultaneously by multiple users or sequentially by different users in the same region.
3) The number of requests $N_{m,f}^{L}$ for content f by user m during the time frame, that a user may access the content with many times. A larger value of $N_{m,f}^{L}$ in fact implies that user m is less interested in that long-form content.

Then, the priority metric is expressed as

$$P^{D}=w_1\sum_{m=1}^{M}N_{m,f}^{d}+w_2\sum_{m=1}^{M}T_f^{L}-w_3\sum_{m=1}^{M}N_{m,f}^{L} \tag{16}$$

where w_1, w_2, and w_3 are weights to adjust the importance of the number of different users' requests, continuous usage time and the number of the same user's requests, respectively. The positive signs on w_1 and w_2 reflect that higher multiuser request counts and longer usage duration indicate greater content popularity. The negative sign on w_3 serves as a penalty term, when a single user repeatedly requests fragmented portions of the same content, it indicates low sustained interest rather than genuine popularity, and thus the caching priority is reduced. Thus, at the beginning of each frame, UAV prioritizes cached contents with higher priority according to the caching

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priority metric, the contents are ranked and cached in UAVs under the caching capacity limit. Denote S_u as the caching capacity of UAV u. $k_u^f \in \{0,1\}$, if content f is cached in UAV u, $k_u^f = 1$, otherwise $k_u^f = 0$

$$\sum_{f=1}^{F} k_u^f l_{m,f}^t \le S_u. \tag{17}$$

As user demands vary over time, caching update interval L_n should be optimized. The caching update time consumption for UAV u is expressed as

$$T_{\text{Update}}^u = \frac{\sum_{f=1}^{F} k_u^f l_{m,f}^t}{R_{u,s}^t} \tag{18}$$

where $R_{u,s}^t$ is the transmission rate between UAV u and BS, and the total caching update time in frame n is

$$T_{\text{Update}}^n = \max\left\{T_{\text{Update}}^u\right\}. \tag{19}$$

Assuming the user m may leave or enter the UAV's coverage in the next time slot, the transmission system is assumed to support air–ground handovers by employing the 3GPP-proposed coordinated multipoint (CoMP) transmission technology, with the dynamic point selection (DPS) mode chosen for cooperative transmission. Since the A2G handover interruption time is typically milliseconds [45], which is negligible compared to the content transmission latency (i.e., seconds), the handover delay is neglected in this work. Due to the dynamic nature of the caching update interval L_n, the transmission delay in frame n of all users T_{Trans}^n is shown as

$$T_{\text{Trans}}^n = \sum_{t=1}^{k} \max\left\{T_{\text{Trans}}^m\right\} \tag{20}$$

where

$$T_{\text{Trans}}^m = \sum_{u=1}^{U} \frac{l_{m,f}^t}{\rho_{m,f}^t R_{m,u}^t + \left(1-\rho_{m,f}^t\right) R_{m,s}^t}. \tag{21}$$

IV. Problem Formulation

In this work, the objective is to minimize the sum of the system transmission delay per unit file size and the caching update delay per unit file size, as

$$T_{\text{tot}} = \sum_{n=1}^{N} \left(\frac{T_{\text{Trans}}^n}{F_n^T} + \frac{T_{\text{Update}}^n}{F_n^U} \right) \tag{22}$$

where F_n^T and F_n^U are the data sizes of total transmission and update data in frame n. To achieve this, the caching update intervals $\boldsymbol{L} = \{L_n \mid n \in \mathcal{N}\}$, the caching deployment $\boldsymbol{K} = \{k_u^f \mid f \in \mathcal{F}, u \in \mathcal{U}\}$, the UAV trajectory planning $\boldsymbol{Q} = \{Q^t \mid t \in \mathcal{T}\}$ and the bandwidth allocation $\boldsymbol{B} = \{b_{m,u}^t, b_{m,s}^t, b_{u,s}^t \mid m \in \mathcal{M}, t \in \mathcal{T}, u \in \mathcal{U}\}$ are jointly optimized. The problem is formulated as

$$\mathbf{P1:} \quad \min_{\{\boldsymbol{L},\boldsymbol{K},\boldsymbol{Q},\boldsymbol{B}\}} \quad T_{\text{tot}} \tag{23}$$

$$\text{s.t.} \quad L_x \in [L_{min}, L_{max}] \tag{24a}$$

$$k_u^f \in \{0,1\} \tag{24b}$$

$$\sum_{f=1}^{F} k_u^f l_{m,f}^t \le S_u \tag{24c}$$

$$v_{\min}\Delta t \le \|Q_u^{t+1} - Q_u^t\| \le v_{\max}\Delta t \tag{24d}$$

$$Q_{Low} \le Q_u^t \le Q_{Up} \tag{24e}$$

$$0 < b_{m,u}^t, \quad b_{m,s}^t, \quad b_{u,s}^t < 1 \tag{24f}$$

$$\sum_{m=1}^{M}\sum_{u=1}^{U} b_{m,u}^t + b_{m,s}^t = 1, \quad \sum_{u=1}^{U} b_{u,s}^t = 1 \tag{24g}$$

where constraint (24a) is the caching update interval constraint, L_{min} is the minimum caching update delay required by UAVs, and L_{max} is the maximum allowable caching interval based on content usability. Equation (24b) represents whether caching is performed. Equation (24c) denotes the caching capacity constraint. Equation (24d) is the UAV speed constraint, (24e) limits the UAV trajectory, (24f) and (24g) ensure that the bandwidth allocation.

P1 is a mixed-integer nonlinear programming (MINLP) problem. Due to the combination of discrete and continuous variables, along with nonconvex objective functions and constraints, it is classified as an NP-hard problem, which cannot be directly solved. Since the dual time scales can be divided into two levels for hierarchical problem-solving, the overall optimization process is decomposed into two subproblems. As shown in Fig. 2, at the beginning of each frame x, considering both the maximum caching usage times and min update delays, the optimization problem is formulated as P2

$$\mathbf{P2:} \quad \min_{\{\boldsymbol{L},\boldsymbol{K}\}} \quad T_{\text{tot}} \quad \text{s.t.} \quad \text{(24a)}, \ \text{(24b)}. \tag{24}$$

After caching update and deployment, at the beginning of each slot, UAVs should move with user requests, adjusting their positions, and bandwidths are allocated accordingly for UAV-to-user or UAV-to-BS communications, then the total transmission capacity for a single time slot R_{Trans} is maximized. The optimization problem is formulated as P3

$$\mathbf{P3:} \quad \max_{\{\boldsymbol{Q},\boldsymbol{B}\}} \quad R_{\text{Trans}}^t \quad \text{s.t.} \quad \text{(24c)} - \text{(24g)} \tag{25}$$

where $R_{\text{Trans}}^t = \sum_{m=1}^{M}(R_{m,u}^t + R_{m,s}^t)$. It is worth noting that in this article, the task processing decision variables $\{\rho_{m,f}^t\}$ are not our analytic target. UAVs which had cached contents, and the corresponding users are within the coverage area, the users link the UAV directly, otherwise, it connects BS. Moreover, the factors F_n^T and F_n^U in (22) are also determined by the time frame decisions and remains constant within each time slot. Maximizing the instantaneous throughput R_{Trans}^t in P3 is consistent with minimizing the average delay per unit content size in P1, i.e., P3 serves as a per-slot surrogate of the global objective P1 conditioned on the time frame decisions.

Based on the sub-problems P2 and P3, considering the dynamic nature of the environment and the sequential decision-making process, HRL is employed to enable effective optimization of time scale variables by dividing the problem into two layers of subproblems, significantly improving the system efficiency.

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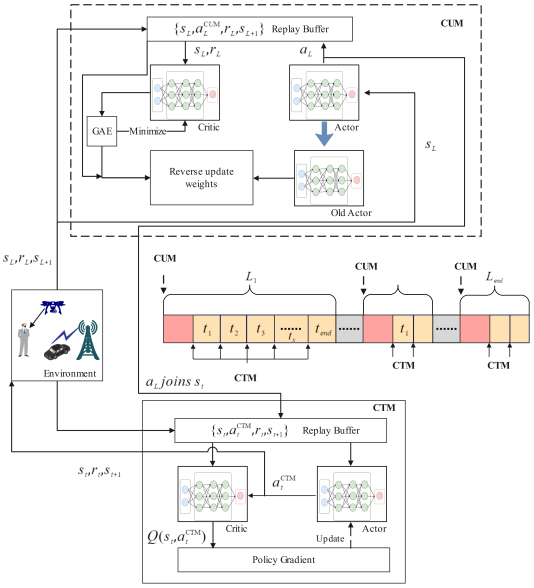


Fig. 3. VD-CURA structural framework.

V. Dual Time Scale Caching Update Algorithm

In this section, we use a HRL-based solution algorithm: VD-CURA. As shown in Fig. 3, combined PPO and deep deterministic policy gradient (DDPG), VD-CURA alternates between optimizing the high-level and low level networks over the time sequence to derive the required subproblem policies. The high level network corresponds to the cache update module (CUM), while the low-level network corresponds to the content transmission module (CTM).

At the beginning of each time frame, users send requests and move dynamically. Simultaneously, the UAV is initialized and checks whether the large time-scale condition is satisfied, which is met once the cumulative time reaches the large time-scale length L_n. If the condition is met, the UAV performs caching update and determines the caching update interval. The caching content selection are based on caching priority, and the update interval is determined by the update time consumption and the maximum allowable content usage time consumption. The high level network selects the optimal values for these two actions to minimize the average delay per unit content size. The UAV hovers during the caching update and, once it's complete, continues to move and allocate bandwidth for UAV–user links in each time slot. The low level network selects appropriate UAV movement and bandwidth allocation ratios to maximize network throughput. This dual time scale structure enables efficient optimization, where the high-level network focuses on minimizing delay by optimizing caching strategies, and the low-level network enhances throughput by optimizing UAV trajectories and resource allocation.

According to PPO, the high level policy $\pi^{\mathrm{CUM}}(g|s;\theta^{\mathrm{CUM}})$ and value network $V^{\mathrm{CUM}}(s;\phi^{\mathrm{CUM}})$ are initialized with random parameters. Similarly, for the proposed DDPG algorithm, the low level policy $\pi^{\mathrm{CTM}}(a|s,g;\theta^{\mathrm{CTM}})$, Q value network $Q^{\mathrm{CTM}}(s,g,a;\phi^{\mathrm{CTM}})$ and their corresponding target networks are initialized. Two experience replay buffers: $\mathcal{D}^{\mathrm{CUM}}$ of PPO and $\mathcal{D}^{\mathrm{CTM}}$ of DDPG are created to store transitions for the CUM and CTM policies respectively. Next, we introduce the details of each step.

A. High Level Optimization via PPO

To solve problem P2, the high level optimization in the VD-CURA framework determines the caching priority list and the next caching update interval for UAV via PPO algorithm, aiming to minimize the average delay per unit content size. The elements of the PPO algorithm are defined as follows.

1) State Space:

$$s_L = \{H_L, \beta_L\} \tag{26}$$

where H_T is the caching expiration rates of UAV, $\beta_L = \{\beta_{L,1}, \beta_{L,2}, \ldots, \beta_{L,F}\}$ denotes the priority scores of the caching content f during time frame T. $\beta_{L,f}$ is the priority score, which is defined by (15).

2) Action Space: The action is defined as

$$a_L^C = \{L_n, \mathbf{K}_n\} \tag{27}$$

where $\mathbf{K}_n = \{k_u^1, k_u^2, \ldots, k_u^F\}$, and k_u^f denotes the content stored in the UAV u. To guarantee that all selected actions comply with the feasibility constraints specified in P2, an action masking mechanism is incorporated into the policy design. Specifically, at each decision step, all infeasible actions—such as those violating cache capacity limits, invalid content indices, or prohibited combinations of $\{L_n, \mathbf{K}_n\}$ are masked out before the policy selects an action. Formally, a binary mask vector $\mathbf{M}_t$ is constructed such that $M_t(i) = 0$ if action i violates any constraint in the problem, and $M_t(i) = 1$ otherwise. The effective policy is then computed as

$$\pi'\left(a_L^C|s_L\right) = \frac{\pi\left(a_L^C|s_L\right) M_t\left(a_L^C\right)}{\sum_{a_L^{C'}} \pi\left(a_L^{C'}|s_L\right) M_t\left(a_L^{C'}\right)}. \tag{28}$$

By restricting the policy to operate within the admissible action subset, VD-CURA ensures that all executed actions inherently satisfy the required optimization constraints.

3) Reward Function: At the beginning of each time frame, the UAV executes caching update actions and determines the update interval. The reward is calculated as the weighted sum of the delay per unit content size for transmission and caching updates, as

$$r_L = \sum_{x=1}^{X} \frac{T_{\mathrm{Trans}}^n}{F_n^T} + \frac{T_{\mathrm{Update}}^n}{F_n^U}. \tag{29}$$

PPO algorithm is applied to solve P2 for the high level strategy, operates on a coarser time scale, selecting a goal a_L in every L steps: delay per unit content size for transmission and caching updates, as

$$a_L \sim \pi^{\mathrm{CUM}}\left(a_L|s_L;\theta^{\mathrm{CUM}}\right). \tag{30}$$

The cumulative reward r_L for the high level policy is calculated by (29). The high level experience $\{s_L, a_L, r_L\}$ is

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stored in $\mathcal{D}^{\mathrm{CUM}}$, and the objective is given by

$$J^{\mathrm{CUM}}(\theta^{\mathrm{CUM}}) = \mathbb{E}\Big[\min\left(r_L(\theta^{\mathrm{CUM}})A_t, \mathrm{clip}(r_L(\theta^{\mathrm{CUM}}), 1-\epsilon, 1+\epsilon)A_t\right)\Big] \quad (31)$$

where $r_L(\theta^{\mathrm{CUM}})$ is the probability ratio, A_t is the estimated advantage, and ϵ is a hyperparameter controlling the clipping range.

B. Low Level Optimization via DDPG

Once the high level network determines the caching priority list and caching update interval, the low-level network executes actions to minimize delays at each time slot via DDPG. The caching priority list influences the movement of UAVs, which follow the caching priority order to move toward user requests. At each time slot, the UAV hovers and updates its state. The elements of the DDPG are defined as follows.

1) State Space:

$$s_t = \{S_u, S_B, B_{u\ell}, x_{u\ell}, y_{u\ell}, L_H, K\} \quad (32)$$

where S_u is the number of tasks assigned to the UAV, S_B is the number of tasks assigned to BS, $B_{u\ell}$ represents the bandwidth ratio between UAV-user links at last time slot t_l, $x_{u\ell}$, $y_{u\ell}$ and $z_{u\ell}$ are the UAV coordinates, L_H and K represent high-level actions which guide action decisions at lower levels.

2) Action Space:

$$a_t = \{v_u, \theta_u, B_u\} \quad (33)$$

where v_u is the UAV movement velocity, θ_u denotes the UAV movement direction, and B_u represents the bandwidth allocated to UAV-user links.

3) Reward Function: At the time slot, UAVs move to new positions, and the system allocates bandwidth for the UAV to optimize the system's total throughput. The reward is defined as

$$r_t = R^t_{\mathrm{Trans}} \quad (34)$$

where R^t_{Trans} is the total transmission rate of the system at time t.

In DDPG, the actor network $\mu(s_t|\theta^{\mu})$ parameterizes the policy, mapping states s to actions a. The critic network $Q(s_t, a_t|\theta^{Q})$ estimates the value of state-action pairs. To enhance training stability, DDPG employs target networks μ' and Q' with parameters $\theta^{\mu'}$ and $\theta^{Q'}$. The target networks are updated using a soft update mechanism, as

$$\theta^{\mu'} \leftarrow \zeta\theta^{\mu} + (1-\zeta)\theta^{\mu'} \quad (35)$$

$$\theta^{Q'} \leftarrow \zeta\theta^{Q} + (1-\zeta)\theta^{Q'} \quad (36)$$

where $\zeta \ll 1$ is a soft update parameter that ensures slow and stable updates of the target networks.

The replay buffer stores experiences in the form of tuples (s_t, a_t, r_t, s_{t+1}). During each update step, a mini-batch of experiences is randomly sampled from the buffer. This approach reduces the correlation between sequential samples and enhances data efficiency. The critic network is trained to minimize the temporal difference (TD) error, defined by the Bellman equation, as

$$L\left(\theta^{Q}\right) = \mathbb{E}_{(s_t,a_t,r_t,s'_t)\sim\mathcal{B}}\left[\left(Q\left(s_t, a_t|\theta^{Q}\right) - y\right)^2\right] \quad (37)$$

where the target value y is given by

$$y = r_t + \gamma Q'\left(s'_t, \mu'\left(s'_t|\theta^{\mu'}\right)|\theta^{Q'}\right). \quad (38)$$

The actor network is updated by maximizing the expected Q value, effectively performing gradient ascent on the policy parameters, as

$$\nabla_{\theta^{\mu}} J \approx \mathbb{E}_{s_t\sim\mathcal{B}}\left[\nabla_a Q\left(s_t, a_t|\theta^{Q}\right)\big|_{a_t=\mu(s_t)} \nabla_{\theta^{\mu}}\mu\left(s_t|\theta^{\mu}\right)\right]. \quad (39)$$

Algorithm 1 HRL-Based VD-CURA

Initialize:
High-level policy network parameters θ^{CUM},
Low-level policy (actor) network parameters θ^{CTM} and value (critic) network parameters ϕ^{CTM},
Replay buffers $\mathcal{D}^{\mathrm{CUM}}$ and $\mathcal{D}^{\mathrm{CTM}}$.
Input: learning rates η_{CUM}, η_{CTM}, discount factor γ, total episodes E_M, number of steps per episode L_{th}, etc.
for episode = 1 to E_M **do**
  Reset environment and get initial state s_0;
  $L \leftarrow 0$, $R \leftarrow 0$;
  while $L \le L_{th}$ **do**
    Observe current state s_T;
    Determine high-level option a_L^{CUM} using CUM policy $\pi_{\theta^{\mathrm{CUM}}}(a \mid s_L)$:
    Given high-level decision a_L^{CUM}, execute low-level control steps;
    for t = 1 to L **(Cache update interval) do**
      Observe current sub-state s_t;
      Select low-level action $a_{t,k}^{\mathrm{CTM}}$ output by CTM's actor network $\mu_{\theta^{\mathrm{CTM}}}$;
      Execute a_t^{CTM}, receive immediate reward r_t and next sub-state s_{t+1};
      Store $\{s_t, a_t^{\mathrm{CTM}}, r_t, s_{t+1}\}$ in low-level replay buffer $\mathcal{D}^{\mathrm{CTM}}$;
      Sample training batch from $\mathcal{D}^{\mathrm{CTM}}$, update CTM's actor-critic network parameters $(\theta^{\mathrm{CTM}}, \phi^{\mathrm{CTM}})$.
    end for
    Calculate the high-level reward r_L by (28)
    Get next high-level environment state s_{L+1}, and store $\{s_L, a_L^{\mathrm{CUM}}, r_L, s_{L+1}\}$ in $\mathcal{D}^{\mathrm{CUM}}$;
    Update high-level network parameters θ^{CUM};
    $R \leftarrow R + r_L$, $t \leftarrow L+1$, $s_L \leftarrow s_{L+1}$;
  end while
end for
Output: Trained parameters θ^{CUM} and $(\theta^{\mathrm{CTM}}, \phi^{\mathrm{CTM}})$.

C. Learning Algorithm of VD-CURA

The VD-CURA algorithm operates alternately between high-level and low-level optimization along the time sequence. As illustrated in the Algorithm 1, the optimization of VD-CURA is formalized into two levels. The high level network makes decisions on the large time scale, while the low level

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network optimizes decisions on the small time scale. In each iteration, decisions for the dual time scales are alternated, enabling the network to learn the optimal parameters. The details are as follows:

1) High Level Network of CUM: The high level network operates at the beginning of each time frame n. Based on the state s_L, it samples an action according to the action probability distribution, serves as the first component of the low level state. At the end of the time sequence, the rewards are obtained.

2) Low Level Network of CTM: The low level network operates at each time slot. It observes the system and UAV states, along with the action received from the high level network, to form the state s_t. Then, the network outputs a deterministic action and receives a low level reward, which is fed back to the high level network.

3) Details of Cooperative Training: Although the low-level DDPG aims to maximize the instantaneous transmission rate R^t_{Trans}, while the high-level PPO optimizes the delay per unit content size, the proposed VD-CURA ensures consistency between the two objectives through the HRL structure. Specifically, the high-level policy determines long-term caching and update decisions, which directly shape the system dynamics encountered by the low-level controller. The low-level DDPG is therefore trained under the constraints imposed by the high-level option, and its immediate rewards are collected and aggregated into the high-level reward signal. As a result, the PPO agent learns to select only those options whose induced low-level behaviors contribute to reducing the overall delay. Conversely, the low-level policy adapts its transmission decisions to the caching patterns and update intervals determined by the high-level layer.

The hierarchical interaction establishes a unidirectional decomposition of the global objective: the high-level network optimizes long-term delay performance, while the low-level network focuses on short-term transmission efficiency. The value functions of these two levels are trained on different time scales but are coupled through the option-induced reward structure, ensuring that low-level rate maximization does not conflict with the high-level delay objective. Instead, the low-level controller operates as a sub-policy conditioned on high-level decisions, and therefore its optimal behavior becomes aligned with the high-level objective through the hierarchical credit assignment mechanism in HRL.

D. Computational Complexity Discussion

Let E_M denote the number of training episodes, L_{th} as the number of slot-level interactions per episode, Γ as the mini-batch size, and $|\theta^{\text{CUM}}|$, $|\phi^{\text{CUM}}|$, $|\theta^{\text{CTM}}|$, $|\phi^{\text{CTM}}|$ are the parameter sizes of the upper-level PPO and lower-level DDPG networks. The dominant cost comes from forward and backward propagation of the actor-critic networks. The overall training complexity of VD-CURA is approximated as

$$\mathcal{O}_{\text{total}} = \mathcal{O}\left(E_M L_{th} \Gamma \left(|\theta^{\text{CTM}}| + |\phi^{\text{CTM}}|\right)\right) + \mathcal{O}\left(E_M \frac{L_{th}}{L_n} \Gamma \left(|\theta^{\text{CUM}}| + |\phi^{\text{CUM}}|\right)\right). \quad (40)$$

TABLE I
DEFAULT PARAMETER SETUP

Parameter	Value
User average speed ($\tilde{v}_m$)	4 m/s
Transmission power of UAV (P_u)	0.01 W
UAV speed range	[0, 40] m/s
Cache storage capacity of UAV	2000 MB
Transmission power of BS (P_s)	0.1 W
Total bandwidth	100 MHz
Noise power spectral density	-174 dBm/Hz
The most total number of content files	40
Content file size range	[80, 120] MB
Number of neurons per layer for PPO network	{64, 64}
Discount factor (γ)	0.99
Number of neurons per layer for DDPG network	{256, 128}
DDPG replay buffer size	400
Soft update rate	0.005
Batch size	300

Since the upper-level policy is updated in each time frame, its update frequency is reduced by a factor of L_n, which lowers the total computation compared with single-timescale joint optimization. In addition, the lower-level action space (i.e., UAV motion and bandwidth ratio) has fixed dimension, so the computational cost increases moderately with user scale.

VI. Simulation Results

In this section, the performance evaluation of the VD-CURA is analyzed through numerical simulations. The system setup includes three UAVs and six users. The users' task requests are contents, and the task computation delay is negligible. The map size is set to 2×2 km, with the ground BS located at the center of the map. The UAV operates within the coverage area of the BS and connects through a wireless backhaul link. The UAV update interval L_n is divided into k equal time slots. As in previous work [46], it is set to 1 s to ensure continuous UAV velocity in each time slot. UAV altitude H is fixed at 50 m.

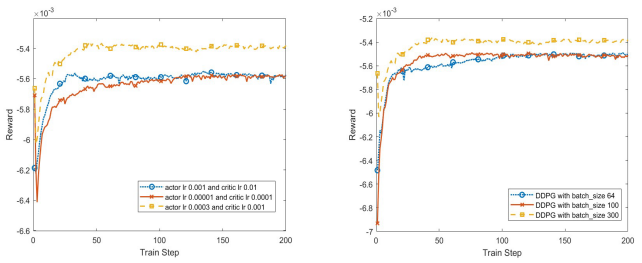
For PPO, Tanh activation function is utilized for hidden layers and the linear activation function is utilized for output layers. For DDPG, *ReLU* activation function is utilized for hidden layers and the linear activation function is also utilized for output layers. Other parameters including hyper parameters of the PPO and DDPG frameworks used throughout the simulations are summarized in Table I [47].

A. Convergence Performance of VD-CURA

In the proposed VD-CURA framework, the HRL algorithm employs PPO for the high-level network and DDPG for the low-level network. Fig. 4 analyze the convergence performance under different settings.

Fig. 4(a) investigates the effect of different PPO learning rate combinations on convergence performance. All three curves exhibit a rapid increase in the initial phase, followed by a stable plateau. It indicates that regardless of the learning rate combination, PPO is capable of learning effective strategies to achieve stable convergence to some extent. Smaller learning rates generally achieve higher final rewards but may require

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(a) (b)

Fig. 4. Convergence of the proposed VD-CURA. (a) PPO convergence performance. (b) DDPG convergence performance.

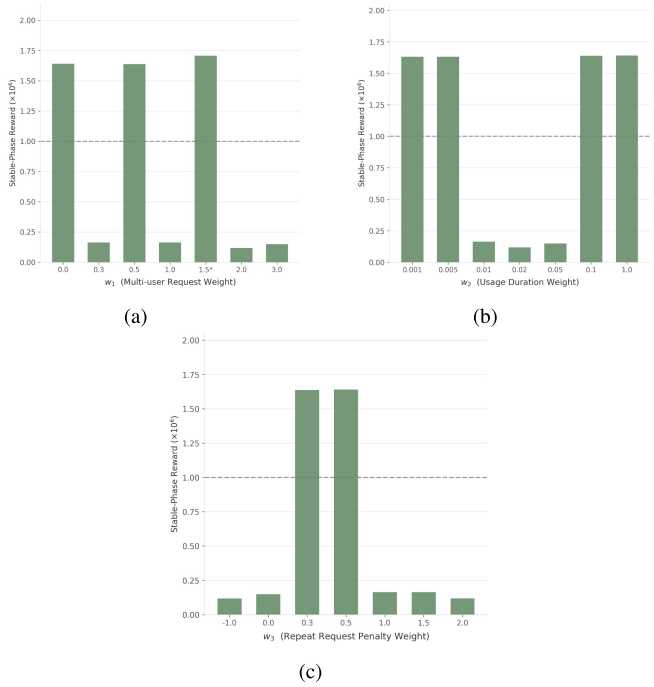
more iterations to fully exploit their advantages. On the other hand, larger or unbalanced learning rates lead to faster initial improvement but often converge to slightly lower final performance. Therefore, the optimal learning rate combination for subsequent experiments is selected as an actor learning rate of 0.0003 and a critic learning rate of 0.001. Fig. 4(b) investigates the impact of different batch sizes. It was observed that all batch size curves show a rapid increase in rewards during the early stages of training, indicating that the algorithm quickly learns preliminary effective strategies. Batch size has a significant influence on the convergence of the DDPG algorithm. Larger batch sizes contribute to faster convergence and higher final rewards but require more computational resources. In contrast, smaller batch sizes provide better exploration capabilities for strategies but may result in slower convergence and instability during the training process.

B. Sensitivity Analysis for Weights of Caching Priority Metric

To determine the optimal weights, we adopt the one-at-a-time (OAT) sensitivity analysis method for the weights of caching priority metric in (16). The experiments are conducted under the standard configuration of $M = 6$ and $N = 3$, each configuration trained for 200 episodes.

1) Sweep Experiment for w_1: Fix $w_2 = 0.01$, $w_3 = 1.0$, and $w_1 \in [0, 3.0]$. As shown in Fig. 5(a), the results exhibit a clear bimodal distribution: configurations with $w_1 \in \{0.0, 0.5, 1.5\}$ converge to the high-performance region, while $w_1 \in \{0.3, 1.0, 2.0, 3.0\}$ fall into low-performance local optima. Among them, $w_1 = 1.5$ achieves the highest stable reward. The main reason is that w_1 yields a moderate priority gap among contents, the cache ranking is sufficiently deterministic for the RL agent to discover patterns. When the gap is small (e.g., $w_1 = 1.0$), the cache ranking fluctuates frequently, making it difficult for the upper-level PPO agent to learn effective update-cycle strategies from unstable cache states. Setting $w_1 = 1.5$ provides a clearer caching decision guidance.

2) Sweep Experiment for w_2: Fix $w_1 = 1.0$, $w_3 = 1.0$, and $w_2 \in [0.001, 1.0]$. As shown in Fig. 5(b), since the baseline configuration ($w_1 = 1.0, w_3 = 1.0$) suffers from the convergence difficulty, $w_2 \in \{0.001, 0.005, 0.1, 1.0\}$ achieve stable rewards. The numerical range of T_f^L (i.e., $[10, 500]$ seconds) is roughly 100 times that of $N_{m,f}^d$ and $N_{m,f}^L$, choosing



(a) (b) (c)

Fig. 5. Weight sensitivity analysis for the caching priority metric. (a) Sensitivity of w_1. (b) Sensitivity of w_2. (c) Sensitivity of w_3.

$w_2 = 0.01$ ensures balanced contributions from all three factors.

3) Sweep Experiment for w_3: Fix $w_1 = 1.0$, $w_2 = 0.01$, and $w_3 \in [-1.0, 2.0]$. As shown in Fig. 5(c), $w_3 \in \{0.3, 0.5\}$ converge successfully, achieving stable rewards, while $w_3 \geq 1.0$ all fail to converge. However, under the already determined optimal $w_1 = 1.5$, the combination with $w_3 = 1.0$ achieves the highest stable reward across all experiments.

According to the three experiments, the final weight configuration is determined as $(w_1, w_2, w_3) = (1.5, 0.01, 1.0)$. This configuration achieves the highest stable reward of 1.707×10^6.

C. Optimize Performance Comparisons

To demonstrate the perform of VD-CURA, the comparative baseline algorithms are described as follows.

1) *Constant Update Frequency (CUF):* The caching update intervals are fixed at 1000.
2) *LRV Cache Replacement Policy (LRVCR):* The cache replacement follows the least recently used (LRV) strategy.
3) *Fixed Bandwidth Allocation (FBA):* The bandwidth allocation is fixed at $b_{m,u}^t = b_{m,s}^t = 0.5$, $b_{u,s}^t = 1/U$.

Fig. 6 compares the average unit delay under different numbers of service users for three baseline algorithms and the proposed VD-CURA. As the number of users increases, the average unit delay optimized by each algorithm remains relatively stable, indicating high robustness of the algorithms.

From Fig. 6, we can find that the proposed VD-CURA performs the best, achieving the lowest average delay across all user numbers. This demonstrates that the VD-CURA outperforms the existing CUF algorithm, as dynamic adjustment of

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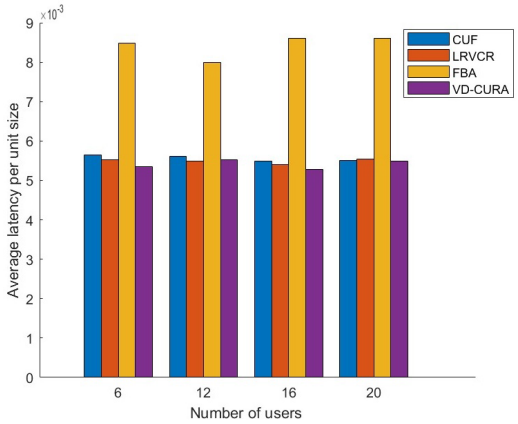


Fig. 6. Comparison of the average unit size of different algorithms.

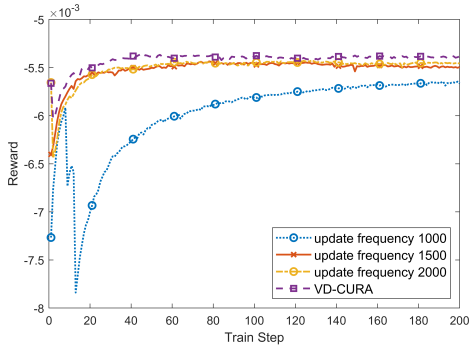


Fig. 7. Comparison of update frequency.

update frequency is more effective than fixed update frequency. Dynamic adjustment better balances the tradeoff between system update delay and transmission delay. In contrast, the FBA algorithm shows the worst performance for all user numbers. This is because the FBA and UAV operations cannot effectively assist in optimizing the system's total delay. LRVCR performs worse than VD-CURA, indicating that the proposed priority metric is superior to the CUF's LRV algorithm. For scenarios with complex content, it is necessary to comprehensively consider the frequency of content requests by a single user, the frequency of content requests by multiple users, and the duration of content usage, rather than solely focusing on request frequency.

Fig. 7 compares the impact of different cache update intervals on the rewards during training.

1) Short Cache Update Intervals: Corresponding to shorter update interval such as 1500, we can find that a rapid increase in rewards during the initial training phase, indicating faster convergence. It demonstrates that frequent cache updates help the model quickly learn and adapt. However, as the training steps increase, the growth of these curves tends to plateau, with final rewards similar to or slightly fluctuating compared to those of longer update intervals. This fluctuation is attributed to the additional computational overhead introduced by frequent cache updates.

2) Long Cache Update Intervals: In contrast, with longer update interval, it exhibits slower reward improvements in the initial phase. As training progresses, these curves steadily increase in rewards, ultimately reaching higher levels. This indicates that longer update interval better maintain the stability of model performance improvement over long-term training, reducing resource consumption and mitigating potential instability caused by frequent updates.

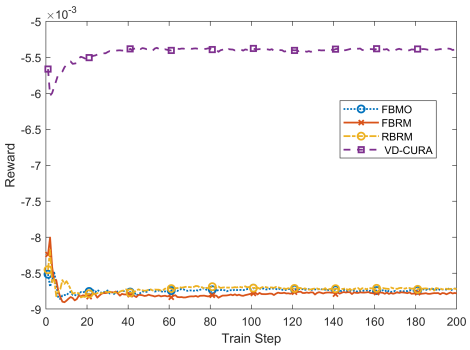


Fig. 8. Comparison of UAV action optimization.

Overall, the results highlight the tradeoff between training efficiency and resource overhead when selecting cache update intervals. The proposed VD-CURA with variable update frequency performs the best.

Fig. 8 compares the system performance under various UAV action strategies. The experiments consider the following four scenarios.

1) *FBMO:* Fixed bandwidth ($b^t_{m,u} = b^t_{m,s} = 0.5$) with UAV autonomous mobility optimization.
2) *FBRM:* Fixed bandwidth ($b^t_{m,u} = b^t_{m,s} = 0.5$) with UAV random mobility.
3) *RBRM:* Random bandwidth with UAV random mobility.
4) *VD-CURA:* Simultaneous optimization of both bandwidth allocation and UAV mobility.

From Fig. 8, except for the proposed VD-CURA algorithm, other three curves converge to suboptimal solutions. The proposed algorithm simultaneously optimizes bandwidth allocation and UAV trajectory, achieves the best performance. The combination of fixed bandwidth and UAV random mobility results in the worst performance. This indicates that when bandwidth is constrained, the improvement of system performance is limited. When the bandwidth is fixed and the UAV can autonomously move and optimize its position, the reward improves. This demonstrates that optimizing UAV positions effectively utilizes limited bandwidth resources, thereby enhancing overall system performance. When both the bandwidth allocation and UAV trajectory are determined randomly, the reward is close to that of fixed bandwidth combined with autonomous UAV mobility optimization. This is because dynamic bandwidth adjustment compensates for the lack of UAV mobility optimization. The proposed VD-CURA achieves the highest reward.

Fig. 9 compares the system performance under different cache replacement strategies. The following three scenarios are considered.

1) *LFUA:* Least frequently used (LFU) algorithm only considers request frequency. It caches content with higher request frequencies into the UAV.

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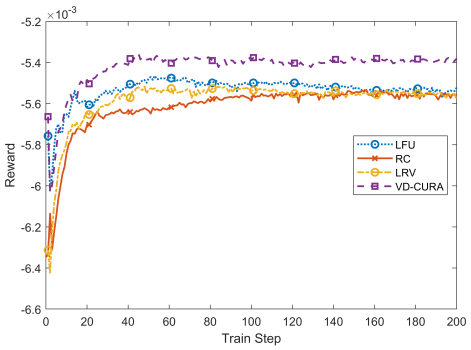


Fig. 9. Comparison of cache replacement strategies.

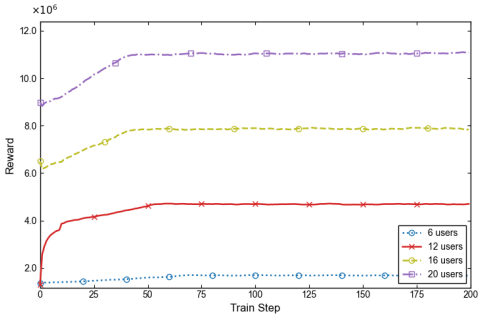


Fig. 10. Scalability of VD-CURA under increasing number of users ($N = 3$).

2) *LRVA:* LRV algorithm replaces cached content based on usage time, prioritizing content that has been used more frequently.
3) *RC:* Random caching (RC) scheme.

It is evident that the proposed VD-CURA algorithm achieves the best performance. In the complex content transmission scenarios, for large files require multiple time slots for completion, a higher request frequency does not indicate a better caching choice. Instead, a comprehensive evaluation of request frequency, content usage duration, and the number of requests from different users is required to determine the caching priority. The proposed VD-CURA achieves the highest reward because it balances the priority of caching content based on multiple factors, ensuring optimal cache utilization in complex scenarios. However, LFUA results a reward lower than that of the proposed one. It indicates that simply prioritizing frequently requested content is suboptimal in scenarios with complex tasks. LRV does not evaluate content popularity, leading to rewards are even lower than those of the LFU. And RC performs the worst due to its random replacement nature. The cache hit rate is unstable and generally low.

D. Scalability Analysis

To evaluate the scalability of VD-CURA beyond the baseline setting, we fix the number of UAVs at $N = 3$ and increase the number of users from 6 to 20. As shown in Fig. 10, all four configurations converge stably within 200 training episodes. As the number of users grows, the stable-phase reward increases accordingly, since increasing number of users generate more requests and thus higher total throughput. Importantly, the convergence speed and stability are well maintained across all scales. All curves reach the plateau within approximately 50–75 training steps and remains stable. This demonstrates that VD-CURA scales gracefully with increasing network size. The hierarchical structure of VD-CURA contributes to the scalability, for the upper-level PPO handles caching decisions on a coarse time scale with a compact state space that grows modestly with user count, while the lower-level DDPG optimizes UAV trajectory and bandwidth allocation per slot with a fixed-dimensional action space regardless of user scale. These properties ensure that the computational overhead of VD-CURA remains manageable as the network scales up.

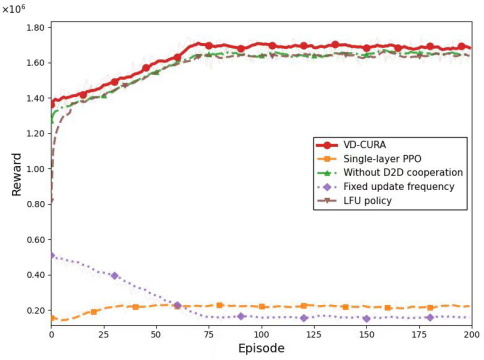


Fig. 11. Training rewards episode for VD-CURA and ablation baselines.

E. Ablation Study

We isolate major design choices by comparing the full VD-CURA against training-stage ablations and flat RL alternatives. The following four scenarios are considered.

1) *Single-Layer PPO:* It uses one flat PPO policy over the joint decision space (i.e., caching, UAV motion, and bandwidth) without the proposed two-timescale scheme.
2) *Without D2D Cooperation:* It disables device-to-device (D2D) assisted content forwarding among ground users that is enabled in the full proposed system when users are in the range.
3) *Fixed Update Frequency:* It keeps the cache update interval fixed while retaining the remaining learning pipeline aligned with VD-CURA.
4) *LFU Policy:* It substitutes the proposed priority metric with the least-frequently-used replacement rule.

Fig. 11 compares the proposed hierarchical design against these ablations. The proposed VD-CURA improves in early training and converges to the highest reward, indicating effective coordination between the upper-level policy and the lower-level execution. Without D2D cooperation follows a similar upward trend but finishes below the proposed method, quantifying the gain from D2D-assisted collaboration. The LFU policy reaches a higher plateau in the late stage, showing that cache replacement has measurable impact yet is not the sole determinant when other components are fixed. By contrast, single-layer PPO remains at a low reward

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with limited improvement, supporting the necessary for the two-timescale hierarchical structure. Fixed update frequency exhibits degraded performance over time and ends at the lowest level, highlighting that the adaptive or learned update timing is critical for stable learning. Overall, the results corroborates that each major component contributes to the final performance, and the full proposed method is preferable for maximizing the long-horizon reward.

VII. Conclusion

This article proposes an efficient various dual time scale caching update and resource allocation scheme to minimize the average system task processing delay per unit content size. It places caching updates on a large time scale to improve content utilization. Meanwhile, it dynamically adjusts UAV trajectories and bandwidth allocation at a small time scale to adapt to user mobility and demand variations. In addition, a new content selection metric is proposed, which comprehensively considers both the duration of content usage and the frequency of user requests for that content type, effectively enhancing cache content utilization. A HRL-based optimization algorithm, VD-CURA is developed. Extensive simulation results demonstrate that the proposed VD-CURA exhibits excellent convergence and effectiveness, optimizing caching strategies in complex task scenarios while significantly reducing system delay and caching content update costs.

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